adani™	STANDARD QUALITY PLAN														Approved by QA
	ITEM: Rolling Shutter							**QP NO**: ADANI / QA / SQP / C -005 **REV.NO**. : 1 **DATE**: 26.08.2013 Page 1 of 6						
SL NO	**COMPONENT/ OPERATION**	**CHARACTERISTICS**	**CATE-GORY OF CHECK#**	**TYPE OF CHECK**	**QUANTUM OF CHECK**			**REFERENCE DOCUMENT**	**ACCEPTANCE NORMS**	**FORMAT OF RECORD**		**INSPECTION AGENCY**			**REMARKS**
					M	**C**	**A**					**M**	**C**	**A**	
1	2	3	4	5	6			7	8	9	D *	**10			11
1	**Raw Material Verification.**	a. **Cold Rolled Steel** Strips used for rolling shutter lath section.	Major	TC Review	100%	100%	100%	IS : 6248 (Latest Revision)	**NOTE 1**	TC	√	P	V	V	**Note 1:** • Conforming to Temper No. 5 Dead soft quality of IS 4030 (LR). • Thickness of the steel strips should not be less than 0.9 mm for shutter up to & including 3.5 m width AND not less than 1.2 mm for shutter above 3.5 m width.
		b. **Mild Steel Sections of Hot Rolled Steel** used for Guide channels , brackets & lock plate.	Major	TC Review	100%	100%	100%	IS : 6248 (Latest Revision)	**NOTE 2**	TC	√	P	V	V	**NOTE 2:** • Thickness > 3.15 mm. • Free from surface defects and edges cleanly sheared. • Conforming to IS 5986 (latest revision)
		c. **Steel Pipes** used for the suspension shaft of the roller.	Major	TC Review	100%	100%	100%	IS : 6248 (Latest Revision)	**NOTE 3**	TC	√	P	V	V	**NOTE 3:** • Heavy duty Pipe. • Conforming to IS 1161 (latest revision)

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SL NO	COMPONENT/ OPERATION	CHARACTERISTICS	CATE-GORY OF CHECK#	TYPE OF CHECK	QUANTUM OF CHECK			REFERENCE DOCUMENT	ACCEPTANCE NORMS	FORMAT OF RECORD		INSPECTION AGENCY			REMARKS
					M	C	A					M	C	A	
1	2	3	4	5	6			7	8	9	D *	**10			11
1	Raw Material Verification.	d. **Cast Iron Casting** used for roller pulley wheels, U – clamps & bevel gears.	Major	TC Review	100%	100%	100%	IS : 6248 (Latest Revision)	**NOTE 4**	TC	√	P	V	V	**NOTE 4:** •Casting shall be free from blow holes, surface defects such as cracks, burns etc. •Conforming to Grade 15 of IS 210 (latest
		e. **Springs** used in the roller for counter balancing the rolling shutter.	Major	TC Review	100%	100%	100%	IS : 6248 (Latest Revision)	**NOTE 5**	TC	√	P	V	V	**Note 5:** •Springs made either from high tensile spring steel wire or flat spring steel strip. •The spring steel wire shall Conforming to Grade 2 of IS 4454 (Part I) (latest revision) •Flat spring steel strip shall be from 0.8 to 1.0 % carbon steel strip,
		f. **Malleable Cast Iron** used for clips.	Major	TC Review	100%	100%	100%	IS : 6248 (Latest Revision)	**NOTE 6**	TC	√	P	V	V	**Note 6:** •Confirm to IS 2108 (latest revision)

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SL NO	**COMPONENT/ OPERATION**	**CHARACTERISTICS**	**CATE-GORY OF CHECK#**	**TYPE OF CHECK**	**QUANTUM OF CHECK**			**REFERENCE DOCUMENT**	**ACCEPTANCE NORMS**	**FORMAT OF RECORD**		**INSPECTION AGENCY**			**REMARKS**
					M	**C**	**A**					**M**	**C**	**A**	
1	2	3	4	5	6			7	8	9	D *	**10			11
1	Raw Material Verification.	g. **Aluminum Alloy Extrusions** for the components of rolling shutters of aluminum.	Major	TC Review	100%	100%	100%	IS : 6248 (Latest Revision)	**NOTE 7**	TC	√	P	V	V	**Note 7:** • Confirm to 53000 (NE 5) or 64430 (HE 30) of IS 733 (latest revision)
2	Fabrication of various parts of rolling shutter.	a. Curtain. b. Lock Plate c. Guide Channels d. Bracket Plate. e. Roller f. Hood Covers g. Gear warms h. Fixing Bolts i. Safety Devices	Major	Physical	100%	100%	Random 10 %	IS : 6248 (Latest Revision)	**Cl. No. 7** FABRICATION of IS : 6248 (Latest Revision)	IIR	√	P	P	V	Fabrication requirement for various parts or rolling shutter laid in IS 6248 needs to be followed by vendor and shall be verified by owner during final inspection.

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SL NO	**COMPONENT/ OPERATION**	**CHARACTERISTICS**	**CATE-GORY OF CHECK#**	**TYPE OF CHECK**	**QUANTUM OF CHECK**			**REFERENCE DOCUMENT**	**ACCEPTANCE NORMS**	**FORMAT OF RECORD**		**INSPECTION AGENCY**			**REMARKS**
					M	**C**	**A**					**M**	**C**	**A**	
1	2	3	4	5	6			7	8	9	D *	**10			11
3	Operation. (Among these three types of rolling shutter operation system, one shall be applicable based on PO requirement.)	a. **Self Coiling** Type Rolling Shutter.	Major	Physical	100%	100%	Random 10%	IS : 6248 (Latest Revision)	**Cl. No. 8.1** FABRICATION of IS : 6248 (Latest Revision)	IIR		P	P	V	
		b. **Gear Operated** Type Rolling Shutter.	Major	Physical	100%	100%	Random 10%	IS : 6248 (Latest Revision)	**Cl. No. 8.2** FABRICATION of IS : 6248 (Latest Revision)	IIR	√	P	P	V	
		c. **Electrically Operated** Rolling Shutter.	Major	Physical	100%	100%	Random 10%	IS : 6248 (Latest Revision)	**Cl. No. 8.3** FABRICATION of IS : 6248 (Latest Revision)	IIR		P	P	V	Electric motor should be operating on 400/440 V, 3 phase, 50 cycles AC supply.
4	Painting	a. Surface Preparation.	Major	Visual	100%	100%	Random 10%	IS : 6248 (Latest Revision)	**NOTE 8**	IIR		P	V	V	**NOTE 8** Surface should be free from rust, oil, grease or any other foreign material prior to start painting.

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					M	C	A					M	C	A	
1	2	3	4	5	6			7	8	9	D*	**10			11
		b. Primer Application (Zinc Phosphate Epoxy Primer)	Major	Measurement	100%	100%	Random 10%	IS : 6248 (Latest Revision)	Two Coats of DFT 75 + 75 micron Total 150 Micron	IIR	√	P	V	V	Primer & Final Paint needs to be carried out at manufacturing shop and touch up shall be carried out by vendor at site after complete installation.
		c. Finished Coat Polyurethane Paint.	Major	Measurement	100%	100%	Random 10%	IS : 6248 (Latest Revision)	One coat of DFT 50 micron.	IIR	√	P	V	V	
5	Dimension	Width, height & thickness of Rolling shutter with accessories.	Major	Measurement	100%	100%	Random 10%	**NOTE 9**	**NOTE 9**	IIR	√	P	V	H	**NOTE 9** Measurement sheet of **Door Openings** jointly measured by Vendor & Owner's Site Representative.

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					M	**C**	**A**					**M**	**C**	**A**	
1	2	3	4	5	6			7	8	9	D*	**10			11
6	Marking & Packing	Marking on Rolling Shutter including it's accessories & packing list.	Major	Physical verification on	100%	100%	100%	IS : 6248 (Latest Revision) and Packing List	IS : 6248 (Latest Revision) and Packing List	Packing List	√	P	P	H	

LEGEND:-

D* Records identified with tick (✓) shall be essentially included by supplier & manufacturer in Quality Documentation package.

** M: Manufacturer / Sub-Supplier, C: Main Supplier, A: **ADANI** or their authorized representative.

Use the following term as appropriate in columns 10.

P: Perform, **V**: verification and **H**: Customer Hold Point to be witnessed and work shall not proceeded till it is witnessed and cleared in writing.

TC: Test Certificate, **TR**: Test Report, **IIR**: Internal Inspection Report